Software Requirements Specification: Ischemia Predictor

Version 1.2 (two models per family) · Ischemia Detection Team · Nov 13, 2026

Revision History

Name	Date	Reason for changes	Version
Ischemia Detection Team	Oct 2026	Initial draft of the SRS	1.0
Ischemia Detection Team	31 Oct 2026	Updated after implementation: dataset and input list fixed, TBD-1 to TBD-4 closed, deviations recorded in Appendix D	1.1
Ischemia Detection Team	13 Nov 2026	Added a second model to each family (gradient boosting, residual MLP, quantum kernel SVM); results, status and deviations in Appendices C and D updated; coverage re-measured (87%)	1.2

1. Introduction

1.1 Purpose

This document specifies the software requirements for Ischemia Predictor, release 1.0 (SRS version 1.2). Ischemia Predictor is a machine-learning-based clinical decision-support system that estimates the likelihood of heart disease from patient clinical data. In this release heart disease is used as a **proxy** for myocardial ischemia (insufficient oxygen-rich blood supply to the heart muscle); the model is not clinically validated for ischemia itself. This SRS covers the complete system: the web client, the web server, and the ML, QML and DL prediction microservices.

1.2 Document Conventions

- Requirements are uniquely identified as FR-xx (functional), NFR-xx (non-functional) and SEC-xx (security).
- "Shall" denotes a mandatory requirement; "should" denotes a desirable requirement.
- Every requirement carries its own priority (High, Medium, Low).
- Each requirement is written to be clear, unambiguous, concise and measurable (verifiable by test, inspection or analysis).
- "TBD" marks information not yet available. Since version 1.1 all TBDs are closed (Appendix C).
- Section structure follows the IEEE-style SRS template by Karl E. Wiegers.
- Where the implementation differs from a requirement, the difference is listed in Appendix D.

1.3 Intended Audience and Reading Suggestions

Reader	Suggested sections
Developers	2, 3, 4, 5, Appendix D
Testers	4, 5 (every requirement is verifiable and maps to the Test Plan)
Architects/designers	2, 3, 5.3, 5.4, Appendix B
Project manager / course instructor	1, 2, 4, Appendix D
Clinical users (reviewers)	1, 2.3, 4

1.4 Product Scope

Ischemia Predictor lets authorized healthcare professionals enter a patient's clinical parameters and receive a prediction of "Ischemia Likely" or "Ischemia Unlikely", with a probability and risk category. The prediction target in the training data is heart disease, used as a proxy for ischemia. Its goals are to (a) help clinicians identify patients who may need further examination, (b) provide fast, consistent and explainable predictions, and (c) keep patient data private and secure. The product is a decision-support tool only and does not replace a doctor's diagnosis.

1.5 References

1. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
2. Wiegers, K. E., Software Requirements Specification Template, 1999.
3. IEEE Std 829-2008, Standard for Software and System Test Documentation (Test Plan, Part 2).
4. IEEE Std 1016-2009, Systems Design – Software Design Descriptions (Architecture and Design, Part 3).
5. OWASP Application Security Verification Standard (ASVS) 4.0.
6. Dataset `heart.csv`: 918 patient records, 11 clinical features, target `HeartDisease` (closes TBD-1). Source and licence: **[team to state the source and licence of the dataset]**.

2. Overall Description

2.1 Product Perspective

Ischemia Predictor is a new, self-contained product with a client–server architecture built from microservices:

- **Web client:** browser-based UI for data entry and result display.
- **Web server:** API gateway handling authentication, validation, preprocessing, orchestration, persistence and logging.
- **Prediction microservices:** independent ML, DL and QML services, each exposing a prediction endpoint to the web server. Each family holds two models: ML (random forest, gradient boosting), DL (scikit-learn MLP, PyTorch residual MLP) and QML (variational circuit classifier, quantum kernel SVM); one validated version per family is active at a time.

- **Database:** stores user accounts, prediction history, model records and audit logs.

The web client talks only to the web server; the web server talks to the microservices over an internal network. The UML use case diagram is in Appendix B.

2.2 Product Functions

- Secure login and role-based access
- Patient data entry and validation
- Data preprocessing identical to model training
- Heart-disease (ischemia proxy) prediction with probability and risk category
- Result explanation, feature-importance display and medical disclaimer
- Reset of entered data
- Prediction history storage and retrieval
- Model version management and update (administrator)
- Audit logging and user management (administrator)

2.3 User Classes and Characteristics

User class	Description	Priority
Clinician (primary)	Doctor, nurse or cardiology technician; moderate computer skill, uses the system frequently to obtain predictions	Highest
Administrator	IT staff; manages users, model versions and audit logs; high technical skill	High
ML Engineer	Deploys and validates new model versions; high technical skill	Medium
Patient	Does not use the system directly; data subject only	Not applicable

2.4 Operating Environment

- **Client:** latest two major versions of Chrome, Firefox, Edge and Safari on Windows 10+, macOS 12+ and Ubuntu 22.04+ (compatibility testing not yet performed; see Appendix D).
- **Server:** Linux (Ubuntu 22.04 LTS) with containerized services (Docker); also runs directly on Windows and macOS for development.
- **Database:** PostgreSQL 15 or later in container deployments; SQLite for local development.
- **Prediction services:** Python 3.10+ (container image: 3.11) with scikit-learn (ML models and the DL MLP), PyTorch (DL residual MLP), a PennyLane simulator (QML models), Flask, SQLAlchemy, PyJWT and bcrypt (closes TBD-2).

2.5 Design and Implementation Constraints

- Each prediction model shall run as an independent microservice behind a REST interface.
- All communication shall use HTTPS/TLS 1.2 or higher, terminated at the deployment ingress or reverse proxy (the Flask development server uses HTTP).

- Patient data handling shall follow HIPAA-style privacy principles: minimum personal data, encryption of stored inputs, audit logging. This prototype uses a public, de-identified dataset and no real patient data (closes TBD-3).
- The preprocessing pipeline used at inference shall be the same versioned artifact used during training.
- Technologies are limited to open-source tools to keep the project cost-free.

2.6 User Documentation

- User and administrator guide: the project README (setup, training, running, roles, tests).
- Every numeric input field shows its unit and valid range in its label.
- A separate online HTML guide and in-app tooltips are not provided in release 1.0 (Appendix D, D16).

2.7 Assumptions and Dependencies

- A validated, trained model is available before release (dependency on the ML team). In this release ml-1.0, ml-gb-1.0 and dl-1.0 are validated; dl-residual-1.0 and both QML models are not.
- Users have a stable internet or intranet connection.
- Input features match those used in training; features not collected at the clinic are out of scope.
- Hosting infrastructure provides at least 99% uptime (assumed).
- The QML service runs on a quantum simulator, not real quantum hardware.
- The dataset is public and de-identified; the label is a proxy and has not been clinically validated for ischemia.

3. External Interface Requirements

3.1 User Interfaces

- **UI-1:** The web client shall provide a login page, a prediction page (patient-data entry form and result), a history page and, for administrators, a management page.
- **UI-2:** Each numeric input field shall display its unit and valid range, and invalid fields shall be highlighted in red with a message adjacent to the field. Categorical inputs are dropdowns.
- **UI-3:** Every result shall display the disclaimer "Decision-support output; not a substitute for professional medical diagnosis."
- **UI-4:** The UI shall be responsive for screen widths from 360 px to 1920 px.

3.2 Hardware Interfaces

No direct hardware interfaces. The system runs on standard client devices (desktop, laptop, tablet) and server hardware. Clinical parameters are entered manually in this release.

3.3 Software Interfaces

Interface	Description
Web server ↔ ML / QML / DL microservices	REST/JSON; input: preprocessed feature vector (11 values), pipeline version and model version; output:

Interface	Description
	class label, probability, feature importances, model version
Web server ↔ Database	SQL through SQLAlchemy; PostgreSQL over TLS in container deployments; stores users, predictions, models and audit logs
Web client ↔ Web server	REST/JSON over HTTPS (HTTP in local development)

3.4 Communications Interfaces

All client–server and server–microservice traffic shall use HTTPS (TLS 1.2 or higher) in a deployed environment. Messages shall be JSON encoded in UTF-8. Authentication tokens shall be sent in the Authorization header, never in URLs.

4. System Features

4.1 User Authentication and Access Control

Priority: High. Only registered users may use the system, with permissions determined by role. User submits credentials; the system verifies them and opens a session, or shows an error.

- **FR-01:** The system shall require a valid username and password before granting access to any patient-data function. (High)
- **FR-02:** The system shall restrict functions by role (Clinician, Administrator, ML Engineer) as defined in Section 5.5. (High)
- **FR-03:** The system shall end a session after 15 minutes of inactivity. (Medium)

4.2 Patient Data Input and Validation

Priority: High. The user enters clinical parameters; the system rejects invalid data. The user fills the form and clicks Predict; the system validates every field and either proceeds or shows field-level errors.

- **FR-04:** The system shall provide input fields for the 11 clinical parameters in the table below (closes TBD-1). (High)
- **FR-05:** The system shall reject a submission if any required field is empty, non-numeric where numeric is expected, or outside its defined clinical range, and shall not call the prediction service. (High)
- **FR-06:** The system shall list every invalid field in a single error response, each with a message stating the valid range, within 1 second of submission. (High)

Field (API name)	Meaning	Accepted values
age	Age in years	1–120
sex	Sex	0 = female, 1 = male

Field (API name)	Meaning	Accepted values
chest_pain_type	Chest pain type	0 = typical angina, 1 = atypical angina, 2 = non-anginal pain, 3 = asymptomatic
resting_bp	Resting blood pressure (mm Hg)	50–250
cholesterol	Serum cholesterol (mg/dl)	50–700
fasting_bs	Fasting blood sugar > 120 mg/dl	0 = no, 1 = yes
resting_ecg	Resting ECG result	0 = normal, 1 = ST-T wave abnormality, 2 = left ventricular hypertrophy
max_hr	Maximum heart rate achieved (bpm)	40–250
exercise_angina	Exercise-induced angina	0 = no, 1 = yes
oldpeak	ST depression induced by exercise	−3 to 7
st_slope	Slope of the peak-exercise ST segment	0 = upsloping, 1 = flat, 2 = downsloping

4.3 Data Preprocessing

Priority: High. Entered data is preprocessed with the same pipeline that was used to train the active model. Valid data is submitted; the pipeline is applied (version checked against the active model) and the feature vector is passed to the model service.

- **FR-07:** The system shall apply the same scaling, encoding and transformation pipeline (same version) that was used to train the active model. The current pipeline (pipeline-1.0) imputes missing values with the median and applies standard scaling; categorical fields are integer codes as in the table above. (High)
- **FR-08:** The system shall reject a prediction request and log an error if the pipeline version does not match the active model version. (Medium)

4.4 Prediction and Result Display

Priority: High. Core feature. Valid data is submitted, preprocessed, sent to the model service and the result is shown.

- **FR-09:** The system shall obtain a prediction from the active validated model of the selected family (ML, QML or DL) for every valid submission. A model that is not validated is not served (FR-22). (High)
- **FR-10:** The system shall display exactly one of "Ischemia Likely" or "Ischemia Unlikely" for each prediction. (High)
- **FR-11:** The system shall display the model's probability as a percentage with one decimal place. (Medium)

- **FR-12:** The system shall classify the result as Low (probability < 30%), Moderate (30% to 70%) or High (> 70%) risk; the thresholds are server settings (`RISK_LOW`, `RISK_HIGH`). (Medium)
- **FR-13:** The system shall display the top 5 features contributing to the prediction, with direction of influence. (Medium)
- **FR-14:** The system shall display the medical disclaimer on every result (UI-3). (High)
- **FR-15:** The system shall allow the clinician to choose the model family (ML, QML or DL) before prediction; the default is the server setting `DEFAULT_MODEL`. The version served within a family is the one activated by an administrator (FR-23). (Low)

4.5 Reset and Error Handling

Priority: High. The user can clear entered data in one step, and service failures are reported without losing the entered data.

- **FR-16:** The system shall provide a "Clear" control that empties all input fields and the displayed result with one click. (Medium)
- **FR-17:** If a prediction microservice is unavailable or does not respond within 10 seconds, the system shall show the message "Prediction service temporarily unavailable, please retry" and shall keep the entered data. (High)
- **FR-18:** The system shall log every error with timestamp, error code and request ID, without logging patient identifiers. (Medium)

4.6 Prediction History

Priority: Medium. Each prediction is stored and can be retrieved later by the clinician who created it. Records are listed newest first, 20 per page.

- **FR-19:** The system shall store each prediction (inputs, result, probability, model version, timestamp, user ID) in the database. Inputs are stored encrypted. (Medium)
- **FR-20:** A clinician shall be able to view only the predictions they created, sorted by newest first, with 20 records per page. (Medium)
- **FR-21:** The system shall allow an administrator to delete a stored record on request, completing deletion within 24 hours (provided through the API). (Low)

4.7 Model Management

Priority: High. Only validated models are served; new model versions are registered by the ML Engineer and activated by the administrator.

- **FR-22:** The system shall use only models marked "validated" in the model registry. (High)
- **FR-23:** The system shall allow an ML Engineer to register a new model version (through the API) and an administrator to activate it without taking the web server offline; activating a validated version deactivates the other versions of the same family. (Medium)
- **FR-24:** The system shall record the model version used with every prediction (see FR-19). (High)

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **NFR-01:** For valid input, the system shall return a prediction within 3 seconds for at least 95% of requests under a load of 50 concurrent users. (High)
- **NFR-02:** The UI shall load the entry form within 2 seconds on a 10 Mbps connection. (Medium)
- **NFR-03:** The system shall support 200 concurrent users with prediction response time increasing by no more than 25% over the 50-user baseline (scalability). (Medium)

5.2 Safety Requirements

- **NFR-04:** Every prediction display shall carry the statement that it is decision-support output and not a medical diagnosis. (High)
- **NFR-05:** The system shall never present a result without its probability and model version in the stored record, so that clinicians can audit it. (Medium)

5.3 Security Requirements

Security objectives

- **SO-1 Confidentiality:** patient data and predictions are accessible only to authenticated, authorized users.
- **SO-2 Integrity:** inputs, models, preprocessing artifacts and stored predictions cannot be altered by unauthorized parties or undetected.
- **SO-3 Availability:** the prediction service remains available and resists abuse such as brute-force login and request flooding.
- **SO-4 Accountability:** every sensitive action is traceable to a user through audit logs.
- **SO-5 Privacy:** personally identifiable information (PII) is collected only when necessary and protected at rest and in transit.

Security requirements

- **SEC-01:** The system shall enforce passwords of at least 10 characters including upper-case, lower-case, digit and symbol, and shall store them only as salted hashes (bcrypt). (SO-1; High)
- **SEC-02:** The system shall lock an account for 15 minutes after 5 consecutive failed logins. (SO-3; High)
- **SEC-03:** The system shall encrypt all data in transit with TLS 1.2 or higher (at the ingress or reverse proxy) and all stored patient inputs with AES-256. (SO-1, SO-5; High)
- **SEC-04:** The system shall enforce role-based access control on every API endpoint and return HTTP 403 for unauthorized access. (SO-1, SO-2; High)
- **SEC-05:** The system shall write an audit log entry (user, action, timestamp, IP) for each login, prediction, deletion, model activation and role change, and retain logs for at least 1 year. (SO-4; Medium)
- **SEC-06:** The system shall sanitize all inputs against SQL injection and cross-site scripting; the model services shall accept only schema-validated JSON. (SO-2; High)
- **SEC-07:** The system shall verify the checksum of a model artifact before loading it and reject any model that fails verification. (SO-2; Medium)

- **SEC-08:** The microservices shall be reachable only from the web server (internal network); direct public access shall be blocked. (SO-1, SO-3; High)
- **SEC-09:** The system shall rate-limit each user to 60 requests per minute. (SO-3; Medium)
- **SEC-10:** The system shall store only the minimum PII required and shall allow a patient identifier to be replaced by a pseudonymous ID. (SO-5; Medium)

5.4 Software Quality Attributes

- **NFR-06 (Accuracy):** The active model of each family shall achieve at least 85% accuracy and at least 85% recall (sensitivity) on the held-out validation set. The targets are retained after validation (closes TBD-4); achieved results are in Appendix D.
- **NFR-07 (Reliability):** The same input with the same model version shall always produce the same output (100% repeatability over 100 repeated runs).
- **NFR-08 (Availability):** The system shall be available at least 99% of the time during a month, excluding planned maintenance.
- **NFR-09 (Usability):** At least 90% of test clinicians shall complete a prediction without assistance in under 2 minutes after a 10-minute introduction.
- **NFR-10 (Maintainability):** Preprocessing, each model service, and the web server shall be separate modules; replacing a model shall require no change to the web client. Unit-test coverage shall be at least 70%.
- **NFR-11 (Compatibility):** The web client shall function correctly on the browsers and operating systems listed in Section 2.4.
- **NFR-12 (Interpretability):** At least the top 5 contributing features shall be shown for 100% of predictions (see FR-13).
- **NFR-13 (Portability):** Each service shall be deployable through Docker on any Linux host with no source changes.

5.5 Business Rules

Function	Clinician	Administrator	ML Engineer
Make prediction	Yes	No	Yes (test only)
View own history	Yes	Yes	No
Manage users	No	Yes	No
Register model	No	No	Yes
Activate model	No	Yes	No
View audit logs	No	Yes	No

- **BR-1:** Only validated models may be used for predictions.
- **BR-2:** Results are advisory; the final clinical decision rests with a qualified clinician.

6. Other Requirements

- **Database:** tables are created by the application (SQLAlchemy). Migration scripts are not provided in release 1.0.
- **Legal:** HIPAA-style principles are followed; no real patient data is used (TBD-3 closed).
- **Internationalization:** English only in release 1.0; SI units only.
- **Reuse:** preprocessing code is shared as a library (`common/`) between training and inference.

Appendix A: Glossary

Term	Meaning
Ischemia	Reduced blood flow and oxygen supply to tissue, here the heart muscle
Heart disease (proxy)	The dataset label `HeartDisease`, used in this release as a stand-in for ischemia
ML / DL / QML	Machine learning / deep learning / quantum machine learning
ECG	Electrocardiogram
ST depression (oldpeak)	ECG measure of ST-segment lowering during exercise, often related to ischemia
RBAC	Role-based access control
PII	Personally identifiable information
TLS	Transport Layer Security
Microservice	Small, independently deployable service with its own interface
FR / NFR / SEC	Functional / non-functional / security requirement

Appendix B: Analysis Models

Use case model (Figure 1 of version 1.0, unchanged). Actors: Clinician, Administrator, ML Engineer, Prediction Services (ML, QML, DL; secondary actor). Use cases: Log In; Enter Patient Data; Validate Data; Get Prediction; View Result and Explanation; Clear Data; View Prediction History; Manage Users; View Audit Logs; Register Model Version; Activate Model Version. Every actor logs in first; Enter Patient Data includes Validate Data, and the Prediction Services take part in Get Prediction. Re-insert the original diagram from version 1.0 here.

Appendix C: To Be Determined List (all closed since version 1.1; resolutions updated in version 1.2)

No.	Item	Resolution
TBD-1	Final dataset and the exact list of input features (FR-04)	Closed: `heart.csv`, 918 records, 11 features listed in FR-04
TBD-2	Final libraries for ML, DL and QML services (Section 2.4)	Closed: scikit-learn (ML models, DL MLP), PyTorch (DL residual MLP), PennyLane simulator (QML models)
TBD-3	Applicable health-data regulation (Sections 2.5, 6)	Closed: HIPAA-style principles; prototype with public de-identified data, no real patient data
TBD-4	Final accuracy and recall targets (NFR-06)	Closed: 85% / 85% retained. ml-1.0, ml-gb-1.0 and dl-1.0 meet them; dl-residual-1.0 (84.8% accuracy) and both QML models do not

Appendix D: Implementation Status and Deviations

D.1 Model results (heart.csv, 80/20 stratified split, seed 42)

Model	Held-out accuracy	Held-out recall	5-fold CV accuracy	ROC-AUC	Status
ML: ml-1.0 (random forest)	90.2%	93.1%	86.2% ± 3.5%	0.927	Validated; active
ML: ml-gb-1.0 (gradient boosting)	89.1%	92.2%	86.0% ± 2.7%	0.926	Validated; not active (can be activated)
DL: dl-1.0 (MLP, scikit-learn)	87.0%	91.2%	85.7% ± 3.3%	0.917	Validated; active
DL: dl-residual-1.0 (residual MLP, PyTorch)	84.8%	85.3%	80.7% ± 3.2%	0.907	Unvalidated (accuracy below 85%); not served
QML: qml-1.0 (4-qubit variational circuit)	78.8%	83.3%	not run	0.863	Unvalidated (below 85%); not served
QML: qml-kernel-1.0 (2-qubit kernel SVM)	76.1%	78.4%	not run	0.750	Unvalidated (below 85%); not served

Cross-validation is the more conservative estimate. Values are from the training run of 13 Nov 2026; the three original models gave the same results as in the 30 Oct run. No QML model is validated, so none is served. Re-check these values against `models/training_report.json` after each training run.

D.2 Requirement status

- **Implemented:** FR-01 to FR-11, FR-13, FR-14, FR-16 to FR-20, FR-22, FR-24, NFR-04, NFR-05, NFR-07, NFR-12, SEC-01, SEC-02, SEC-04, SEC-06, SEC-07.
- **Implemented with limitations:** FR-09 and FR-15 (QML unavailable), FR-12, FR-17, FR-21, FR-23, NFR-06 (ML and DL only), NFR-10 (coverage measured at 87% overall: gateway 81%, model service 82%, preprocessing 100%, feature definitions 44%; 32 tests pass), NFR-13, SEC-03, SEC-05, SEC-08, SEC-09, SEC-10.
- **Not verified:** NFR-01, NFR-02, NFR-03, NFR-08, NFR-09, NFR-11.

D.3 Deviations from version 1.0

#	Area	Version 1.0 said	What was built
D1	Dataset (TBD-1)	UCI heart disease, TBD	`data/heart.csv`, 918 records; 80/20 stratified split, seed 42
D2	Inputs (FR-04)	9 fields	11 fields (adds fasting blood sugar and ST slope); categories are integer codes; cholesterol and resting BP of 0 are treated as missing and imputed
D3	Target	Myocardial ischemia	`HeartDisease` is a proxy; the UI still says "Ischemia Likely/Unlikely" (FR-10)
D4	DL service (TBD-2)	TensorFlow/PyTorch	scikit-learn MLP (dl-1.0) and a PyTorch residual MLP (dl-residual-1.0)
D5	QML service	Quantum SDK, 85% target	PennyLane simulator: 4-qubit variational classifier (78.8% accuracy, 83.3% recall) and 2-qubit quantum kernel SVM (76.1%, 78.4%); both unvalidated and not served (BR-1, FR-22)
D6	Explanation (FR-13)	Top-5 features	Leave-one-out attribution for the original models; the new models use their own local methods: gradient-boosting importance weighted by the effect of ablating each feature, gradient times input for the residual MLP, and feature ablation for the quantum kernel SVM (approximate, not SHAP; it attributes only its 2 selected features). Internal feature names are shown

#	Area	Version 1.0 said	What was built
D7	TLS (SEC-03)	HTTPS/TLS 1.2+	Development server uses HTTP; TLS expected at ingress or reverse proxy; stored inputs encrypted with AES-256-GCM
D8	Kubernetes (SAD 2.5)	Orchestrated by Kubernetes	Dockerfile and Docker Compose only; only the web port is published (SEC-08 holds in Compose)
D9	Database	PostgreSQL with migrations	SQLite locally, PostgreSQL via `DATABASE_URL` in Compose; no migration scripts
D10	Config (FR-12, FR-15)	Administrator changes thresholds and default model	Server settings `RISK_LOW`, `RISK_HIGH`, `DEFAULT_MODEL`
D11	Rate limit (SEC-09)	60 per minute per user	In-memory, per process, authenticated endpoints only
D12	Pseudonymisation (SEC-10)	Replace patient identifier with pseudonymous ID	`patient_ref` auto-generated when omitted; no replace function
D13	Audit log (SEC-05)	Retained 1 year	Login, prediction, deletion, model activation and user creation are logged; nothing is purged and no retention job exists
D14	Admin functions	Admin UI	Delete prediction (FR-21) and register model (FR-23) are API-only; no change-role call
D15	Error handling (SAD 3.4)	Offer another model when one is down	Entered data is kept; the user selects another model manually
D16	UI (UI-1, 2.6)	Separate result page, tooltips, HTML user guide	Result on the prediction page; units and ranges in labels; user guide is the README
D17	Verification	NFR-01, 02, 03, 08, 09, 11	Not measured (load, browser and usability tests not run)
D18	Retraining	Not specified	Version names are reused; the gateway imports only new versions at start-up, so after retraining delete instance/ischemia.db (then recreate users) or rename versions The new versions ml-gb-1.0, dl-residual-1.0 and qml-kernel-1.0 are imported at the next

#	Area	Version 1.0 said	What was built
			start-up.`instance/ischemia.db` (then recreate users) or rename versions
D19	`create_user` script	Password policy (SEC-01)	The script skips the policy check; the admin API enforces it
D20	Internal API	`/predict`, `/health`	Also `GET /info`, `latency_ms`, NaN/Infinity rejection, 16 KB body limit
D21	Model variants (FR-15, FR-23)	`Not specified`, `/health`	Two models per family. Users choose only the family; an administrator activates one validated version per family (PUT /api/v1/models/<id>/activate). Only ml-1.0 and dl-1.0 are active; QML has no active model`GET /info`, `latency_ms`, NaN/Infinity rejection, 16 KB body limit
D22	Dependencies (2.4)	`Versions not stated`, `/health`	Installed: scikit-learn 1.7.2, PyTorch 2.14.1, PennyLane 0.42.3. requirements.txt does not pin versions; Docker image uses Python 3.11`GET /info`, `latency_ms`, NaN/Infinity rejection, 16 KB body limit
D23	Ports (README)	`Service ports 5001 to 5003`, `/health`	README local run uses 5001, 5002 and 5003; Docker Compose uses 5001 inside each container and publishes only the web port 5000`GET /info`, `latency_ms`, NaN/Infinity rejection, 16 KB body limit